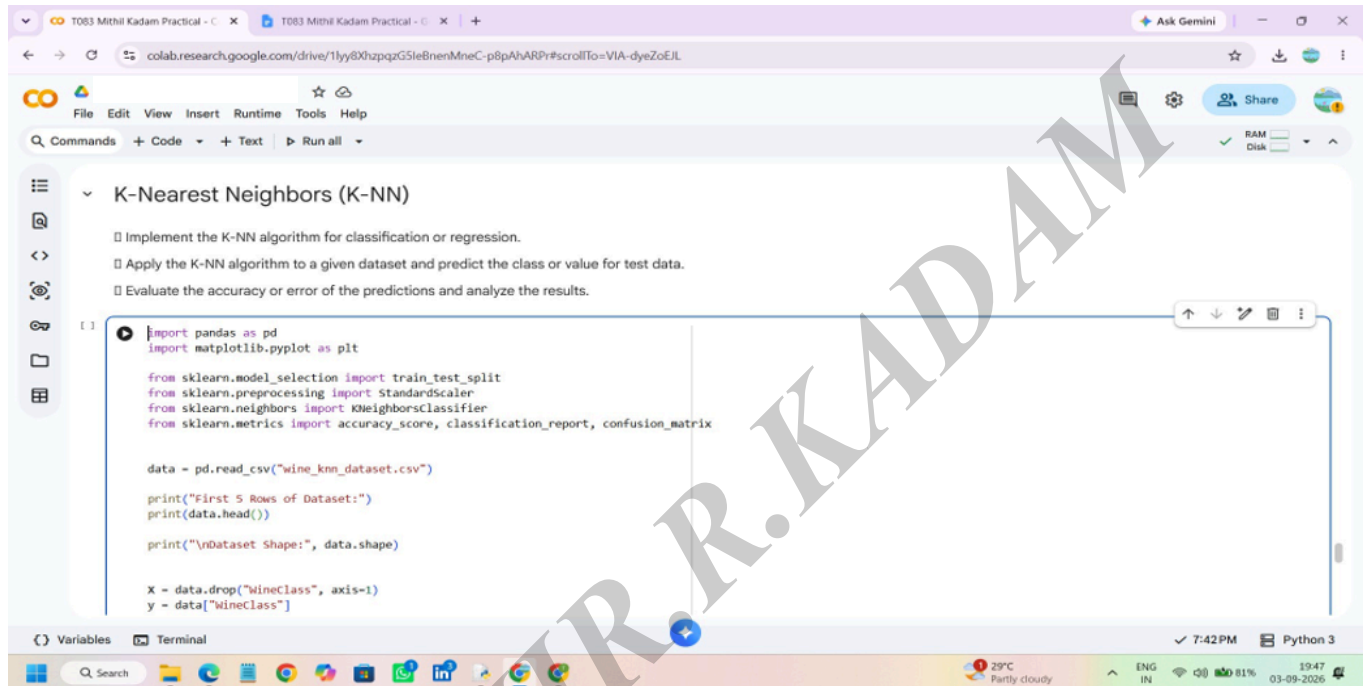


Aim:- K-Nearest Neighbors (K-NN)

Implement the K-NN algorithm for classification or regression.

Apply the K-NN algorithm to a given dataset and predict the class or value for test data.

Evaluate the accuracy or error of the predictions and analyze the results.



```
import pandas as pd
import matplotlib.pyplot as plt

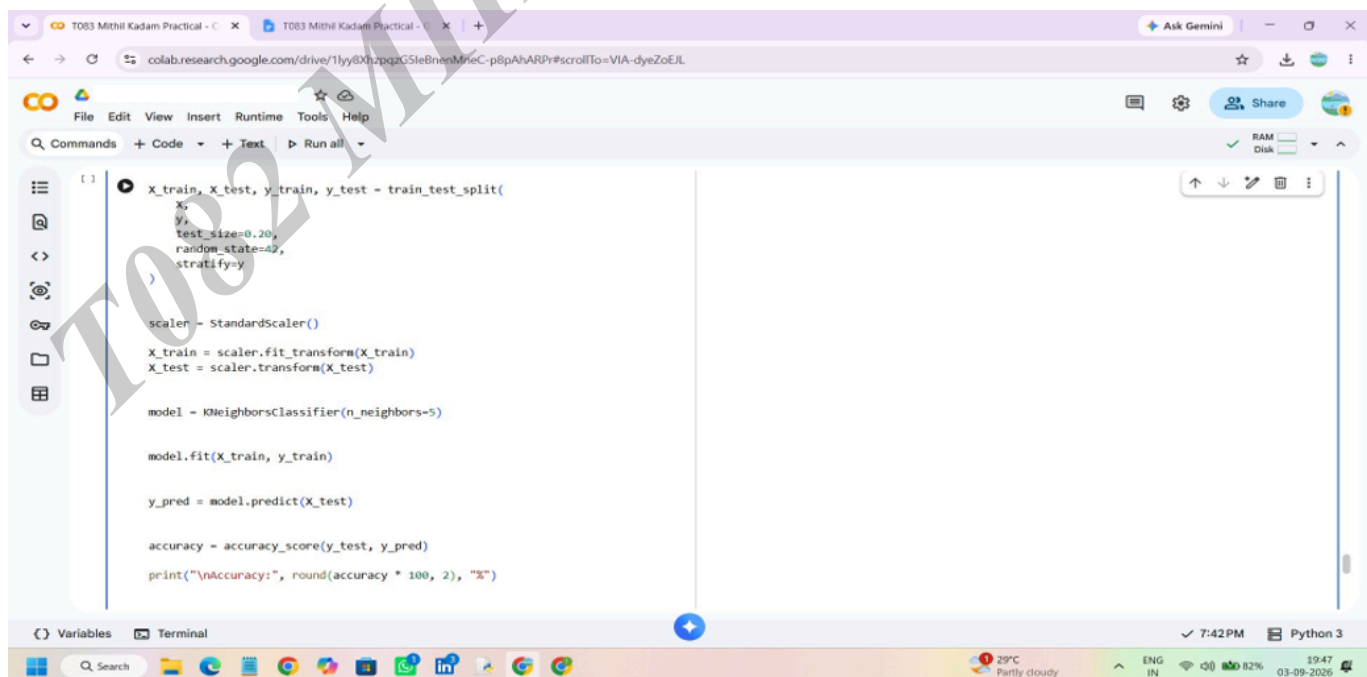
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

data = pd.read_csv("wine_knn_dataset.csv")

print("First 5 Rows of Dataset:")
print(data.head())

print("\nDataset Shape:", data.shape)

X = data.drop("WineClass", axis=1)
y = data["WineClass"]
```



```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

model = KNeighborsClassifier(n_neighbors=5)

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)

print("\nAccuracy:", round(accuracy * 100, 2), "%")
```

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```

print("\nActual vs Predicted Values:")
results = pd.DataFrame({
    "Actual": y_test.values,
    "Predicted": y_pred
})
print(results.head(10))

print("\nClassification Report:")
print(classification_report(y_test, y_pred))

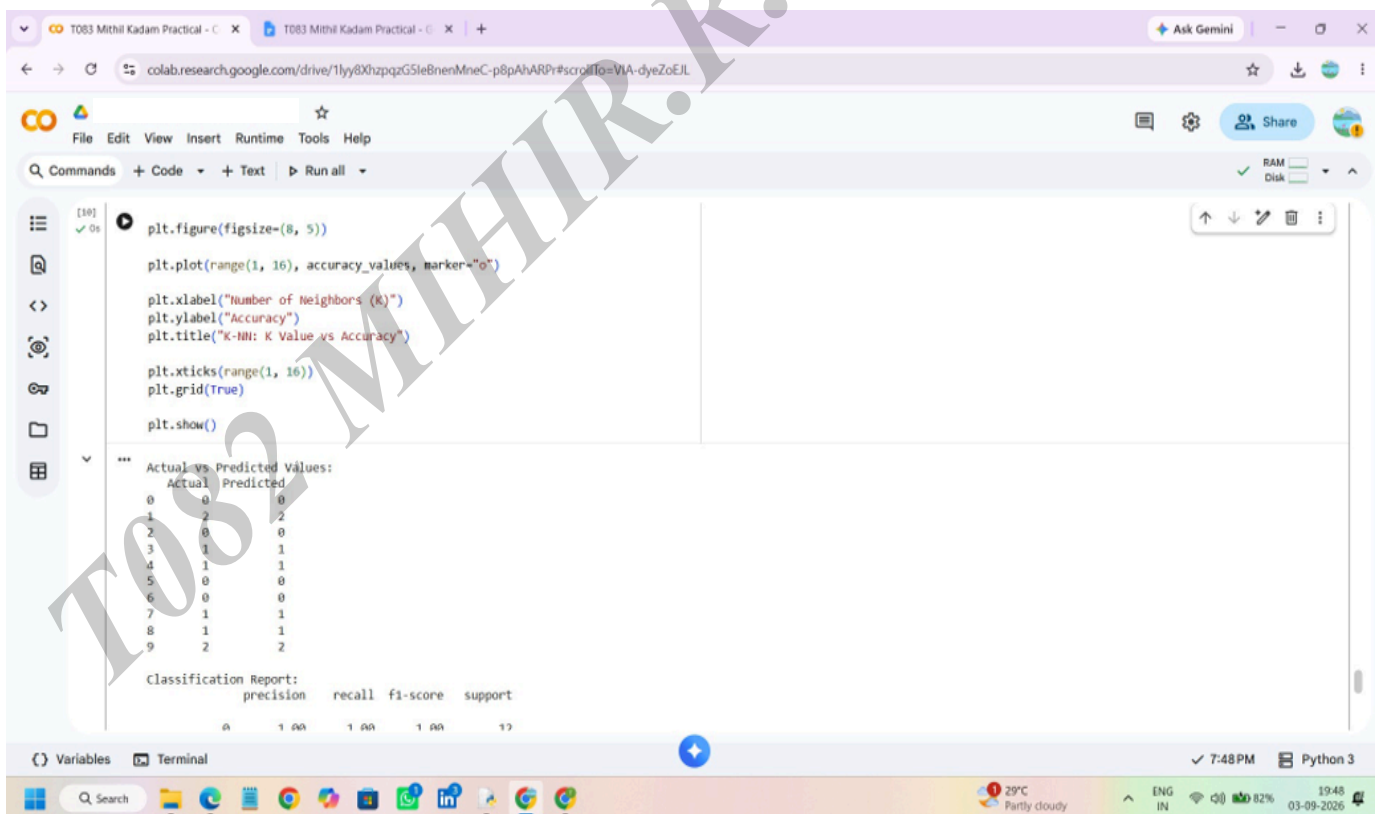
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

accuracy_values = []
for k in range(1, 16):
    knn = KNeighborsClassifier(n_neighbors=k)
    knn.fit(X_train, y_train)

    prediction = knn.predict(X_test)

    score = accuracy_score(y_test, prediction)
    accuracy_values.append(score)

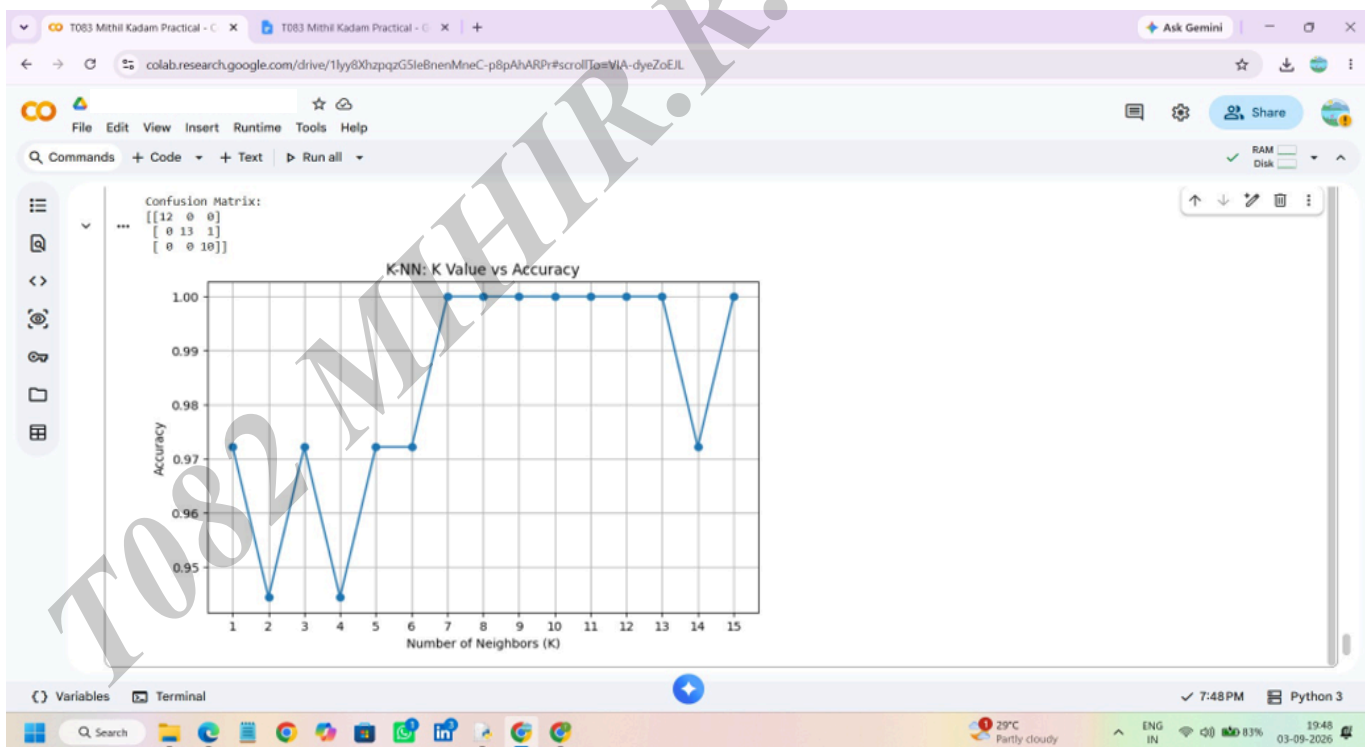
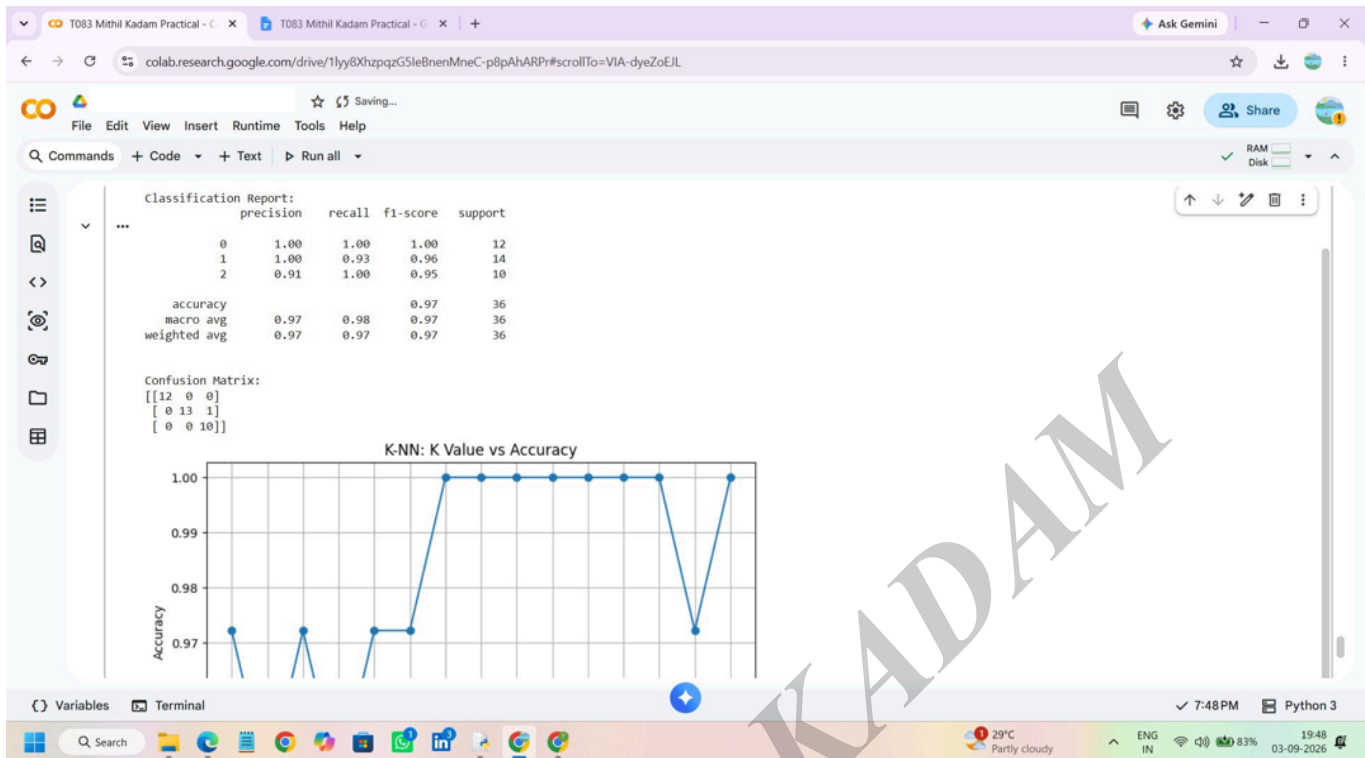
```



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